UNIT 5
INTRODUCTION TO MS EXCEL

Learning Objectives

- Explain MS Excel's purpose and key features.
- Perform basic tasks like data entry, formulas, and chart creation.
- Identify and use key components of the Excel interface.
- Apply Excel for practical tasks like lists, budgets, and charts.
- Navigate rows, columns, and sheets effectively.

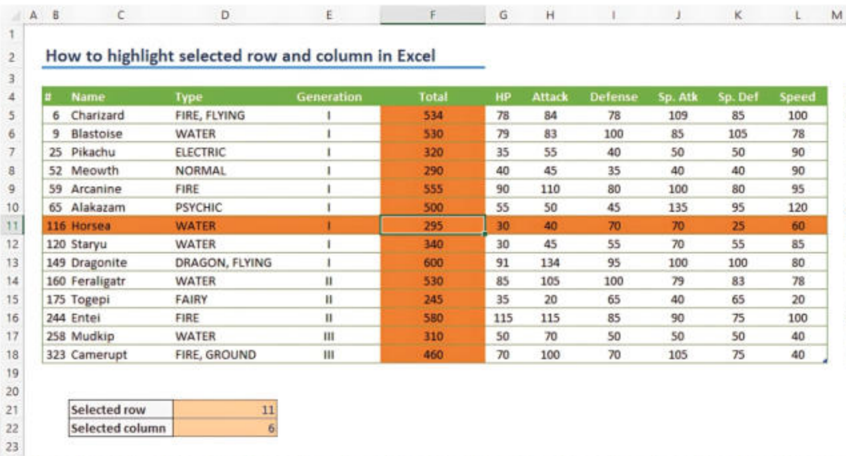
WHAT IS MS EXCEL?

MS Excel is a simple tool that helps you organize information using a table made up of rows and columns. You can use it to make lists, add numbers, and create small charts.

Key Features of MS Excel:

1. Spreadsheet Layout:

- MS Excel has boxes called "cells" arranged in rows (numbers) and columns (letters).
- **Example:** You can use these boxes to make a list of your friends' names or their favourite colours.



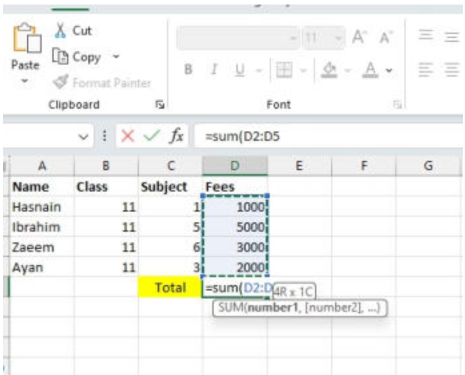
How to highlight selected row and column in Excel

#	Name	Type	Generation	Total	HP	Attack	Defense	Sp. Atk	Sp. Def	Speed
6	Charizard	FIRE, FLYING	I	534	78	84	78	109	85	100
9	Blastoise	WATER	I	530	79	83	100	85	105	78
25	Pikachu	ELECTRIC	I	320	35	55	40	50	50	90
52	Meowth	NORMAL	I	290	40	45	35	40	40	90
59	Arcanine	FIRE	I	555	90	110	80	100	80	95
65	Alakazam	PSYCHIC	I	500	55	50	45	135	95	120
116	Horsea	WATER	I	295	30	40	70	70	25	60
120	Staryu	WATER	I	340	30	45	55	70	55	85
149	Dragonite	DRAGON, FLYING	I	600	91	134	95	100	100	80
160	Feraligatr	WATER	II	530	85	105	100	79	83	78
175	Togepi	FAIRY	II	245	35	20	65	40	65	20
244	Entei	FIRE	II	580	115	115	85	90	75	100
258	Mudkip	WATER	III	310	50	70	50	50	50	40
323	Camerupt	FIRE, GROUND	III	460	70	100	70	105	75	40

Selected row	11
Selected column	6

2. Data Entry:

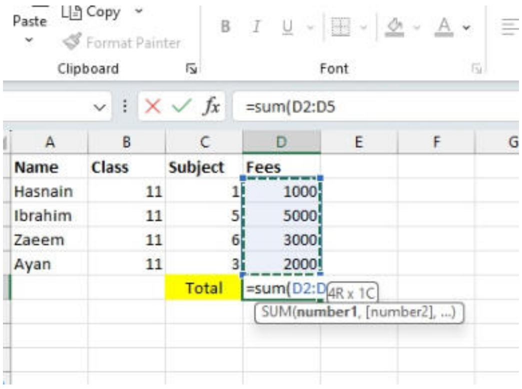
- You can type words, numbers, or even simple maths problems into the cells.
- **Example:** You can add numbers like 1000, 5000,6000,3000 in a cell, and Excel can do the maths for you.



Name	Class	Subject	Fees
Hasnain	11	1	1000
Ibrahim	11	5	5000
Zaeem	11	6	3000
Ayan	11	3	2000
		Total	=sum(D2:D

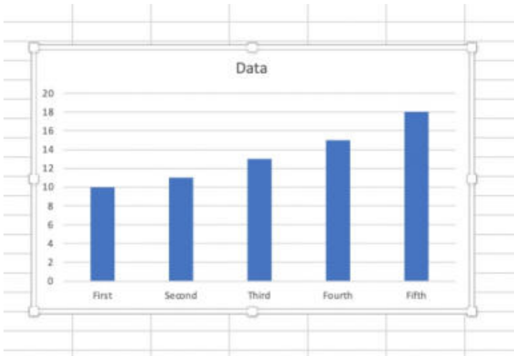
3. Simple Formulas:

- Excel can do maths like adding numbers. One simple formula is =SUM to add numbers.
- **Example:** Type =SUM(D2:D5) in D6 to add the total fees of students as shown in the figure.

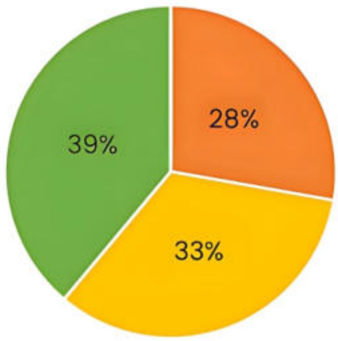


4. Creating Charts:

- Excel can make charts from your data to help you see information in a fun way. Below are two types of charts you can create in Excel.



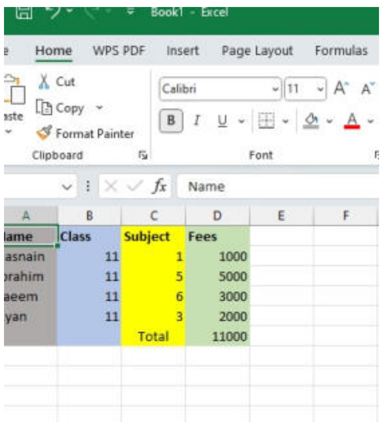
Column chart



Pie chart

5. Formatting Your Work:

- You can change the colours of the cells, make the text bold, or make it look pretty.



APPLICATIONS OF MS EXCEL:

• **Class Lists:** You can create a list of your classmates' names and favourite subjects.
• **Simple Budgets:** You can track how much money you have saved and what you want to buy.
• **Charts for Fun Projects:** You can create a chart to show how many candies you and your friends like!

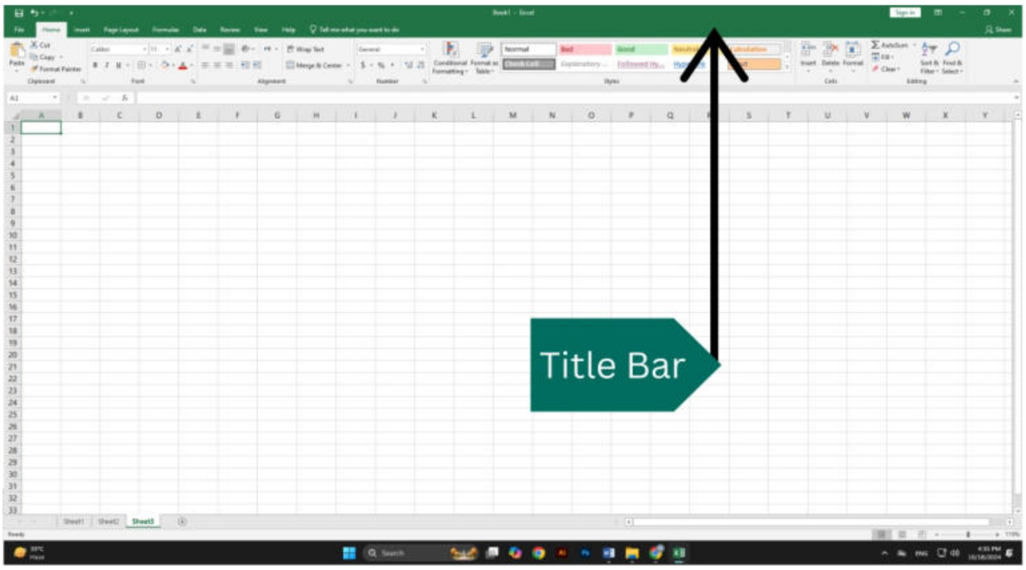
MS Excel is an easy tool that helps you organize information and do simple math. By learning Excel, you can create lists, do calculations, and even make colourful charts!

INTERFACE OF MS EXCEL SHEET

The interface of MS Excel is designed to help users easily access its various features and tools. Below, we will explore the main components of the Excel interface:

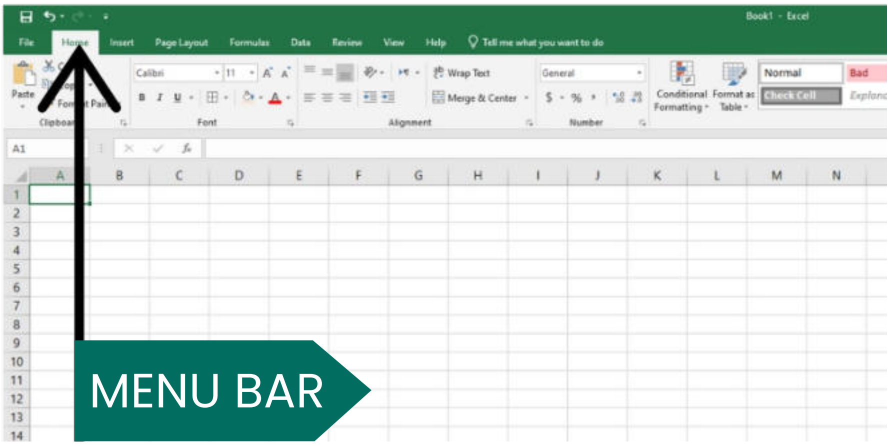
1. Title Bar:

The title bar is located at the very top of the Excel window. It displays the name of the file you are working on (e.g., "Book1 - Excel") and the name of the application ("Microsoft Excel").



2. Menu Bar/Ribbon:

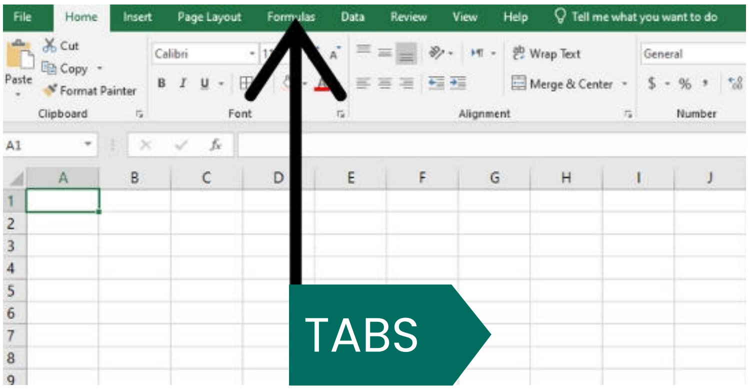
The Ribbon contains different tabs (like Home, Insert, Page Layout, etc.) that group related tools and features. Each tab contains specific commands that you can use in your worksheet.



3. Tabs:

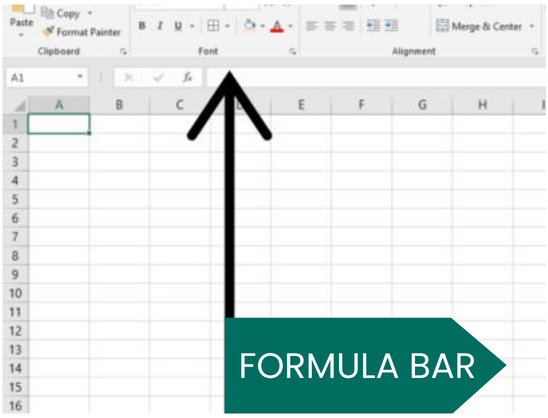
Common tabs include:

- **Home:** Contains commands for formatting text, copying, pasting, and other basic tasks.
- **Insert:** Allows you to add charts, pictures, tables, and other elements to your spreadsheet.
- **Page Layout:** Used for adjusting the overall appearance of the worksheet, including margins and orientation.
- **Formulas:** Contains functions and formulas for calculations.
- **Data:** Used to manage and analyze data, such as sorting and filtering.
- **Review:** Offers tools for spell checking and comments.
- **View:** Changes how you view your worksheet (e.g., zooming in or out, and freezing panes).



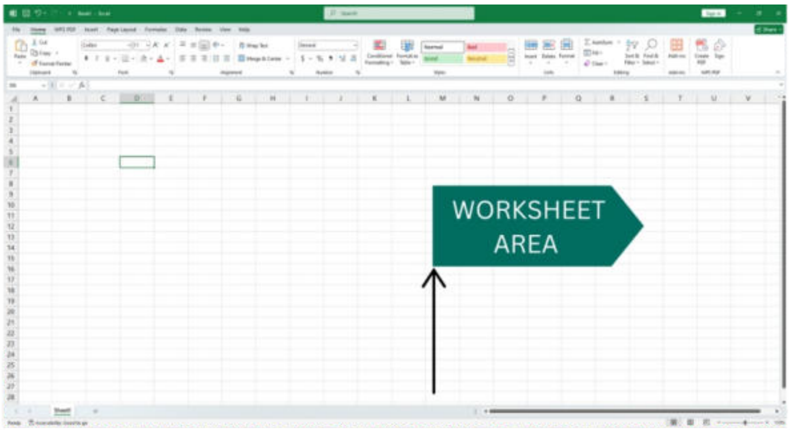
4. Formula Bar:

• Located below the Ribbon, the formula bar shows the contents of the currently selected cell. You can enter or edit data or formulas directly in this bar.



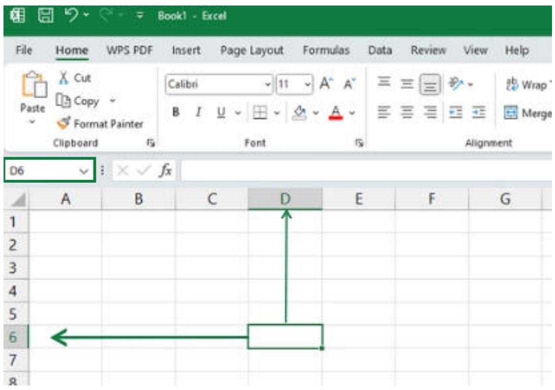
5. Worksheet Area:

• The main part of the interface where you enter and view your data. This area is divided into cells, which are organized in rows (numbered) and columns (lettered).



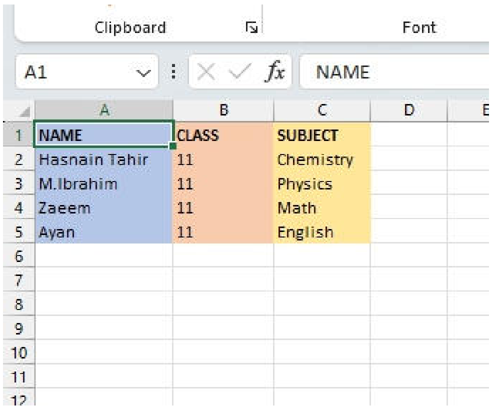
6. Cells:

• Each cell can hold data, formulas, or functions. When you click on a cell, its address (e.g., A1, B2) is displayed in the name box, which is located to the left of the formula bar.



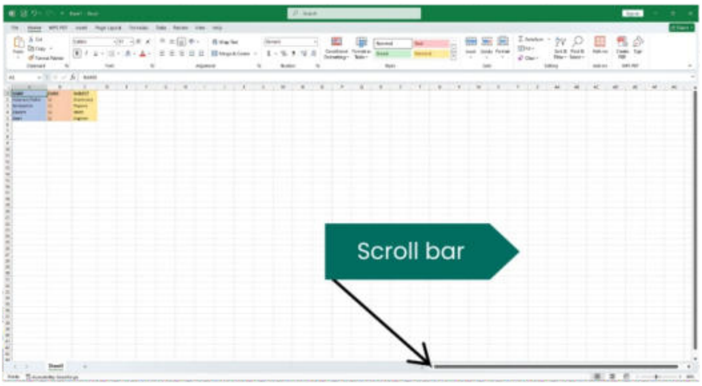
7. Row and Column Headers:

• The row headers (numbers) and column headers (letters) allow you to identify specific cells. You can click on a header to select an entire row or column.



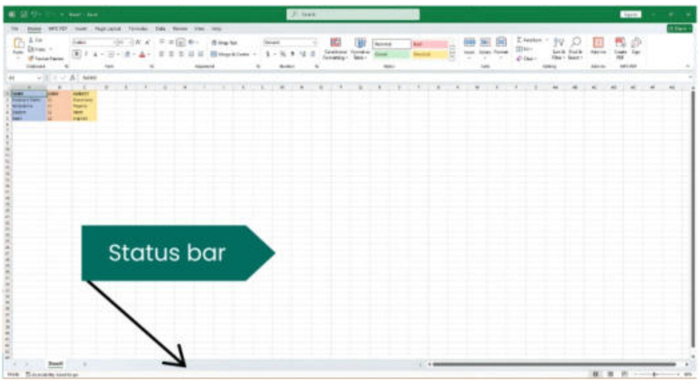
8. Scroll Bars:

• Vertical and horizontal scroll bars located on the right and bottom of the window let you navigate through larger worksheets that extend beyond the visible area.



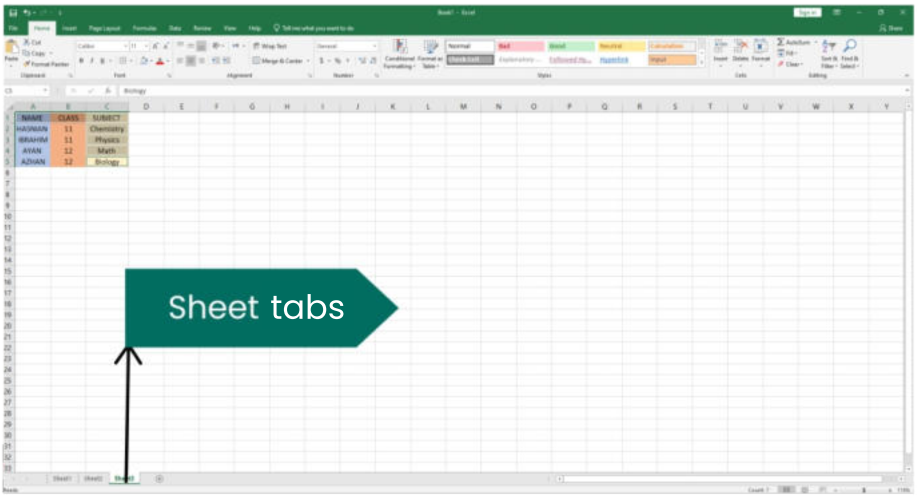
9. Status Bar:

• The status bar at the bottom of the Excel window provides information about the current mode (like "Ready" or "Enter"), and it can also display calculations, like the average or sum of selected cells.



10. Sheet Tabs:

• At the bottom of the window, sheet tabs allow you to switch between different worksheets within the same Excel file. You can add or rename sheets as needed.



Understanding the MS Excel interface is essential for effectively using the software. By familiarizing yourself with its components, you will be better prepared to enter data, perform calculations, and create visually appealing reports.

EXERCISE 1

Choose the correct answer.

1. What is the purpose of a cell in MS Excel?

A. To hold data, formulas, or functions
B. To change the colour of the worksheet
C. To display the current file name
D. To add images to the spreadsheet

2. Which feature in MS Excel is used to perform calculations like addition?

A. Chart
B. Formula Bar
C. =SUM formula
D. Status Bar

3. Where is the Title Bar located in MS Excel?

A. At the bottom of the screen
B. Below the Ribbon
C. At the very top of the screen
D. Next to the Sheet Tabs

4. Which of the following tabs is used to insert charts and images?

A. Home
B. View
C. Insert
D. Data

5. What does the Status Bar display?

A. File name and application name
B. Calculations like sum or average of selected cells
C. Tabs for different sheets
D. Data entries in cells

EXERCISE 2

Fill in the blanks.

1. The ______________ bar at the top of MS Excel displays the file name and application name.

2. Cells are organized in rows, labelled by ______________, and columns, labelled by ______________.

3. To add numbers in MS Excel, you can use the ____________ formula.

4. The ______________ tabs allow you to switch between multiple worksheets within one Excel file.

5. The ______________ bar shows the data or formula currently in the selected cell.

EXERCISE 3

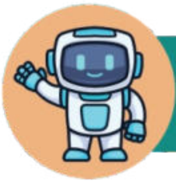
Mark "True" or "False".

	TRUE	FALSE
The Ribbon in MS Excel contains different tabs, each with specific commands.		
The =SUM formula is used to change the colour of cells.		
The Worksheet Area is where you enter data into cells.		
Row headers are labelled with letters, and column headers are labelled with numbers.		
The Formula Bar is used to edit data or formulas in a cell.		

EXERCISE 4

Match the following terms with their descriptions

Term	Description
Title Bar	Contains tabs with commands for formatting and inserting items
Ribbon	Shows the contents of the selected cell
Formula Bar	Shows calculations like sum or average of selected cells
Sheet Tabs	Allows you to navigate larger worksheets that extend beyond the screen
Status Bar	Allows you to switch between worksheets within the same file
Scroll Bars	Displays the file name and Excel application name



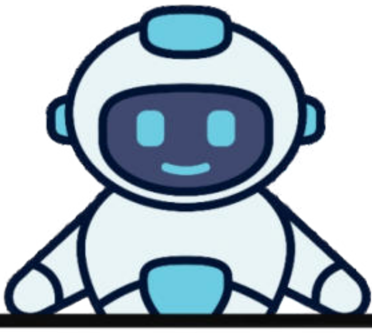
EXERCISE 5

Unscramble the following terms related to MS Excel formatting features.

BRIBON	
ALUMFRO ABR	
HSEET TASB	
LLSCE	
SRWKOEHTE REAA	

LAB ACTIVITY

CREATE A PERSONAL CONTACT LIST

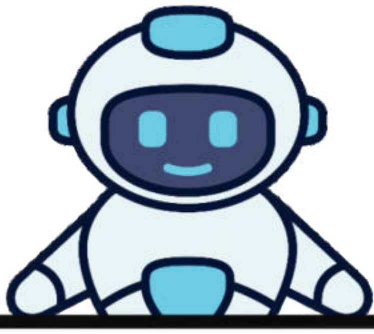


Create a list of your friends or family members' names, phone numbers, and favourite colours in Excel.

- Open Excel and create a new spreadsheet.
- Label the first row with headers: "Name," "Phone Number," and "Favourite Colour."
- Fill in the rows with information about 5-10 friends or family members.

LAB ACTIVITY

FAVOURITE FRUIT SURVEY

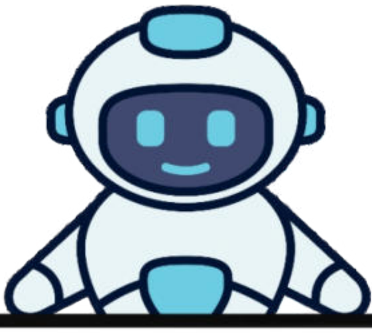


Collect data on your classmates' favourite fruits and create a chart.

- Conduct a class survey asking each student their favourite fruit.
- Enter the data into an Excel spreadsheet with two columns: "Name" and "Favourite Fruit."
- After entering the data, you can create a pie chart to visualize the results.

LAB ACTIVITY

CLASS ATTENDANCE TRACKER

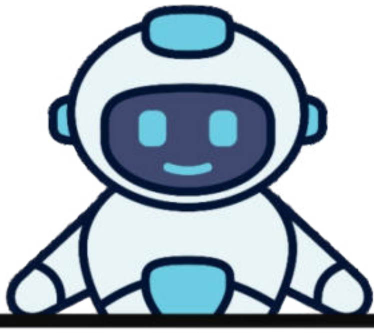


Create a simple attendance tracker for your class.

- Open Excel and label columns as "Date," "Student Name," and "Present/Absent."
- Fill in attendance for a week, marking "P" for present and "A" for absent.
- You can then calculate the total number of present and absent days using the =SUM formula.

LAB ACTIVITY

CREATE A SIMPLE QUIZ



Create a quiz using Excel where classmates can answer questions.

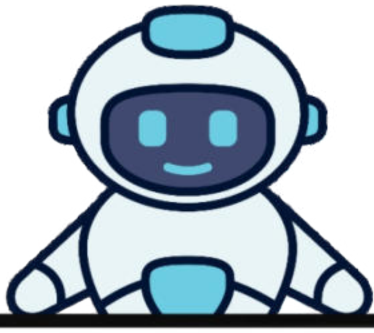
• Open a new Excel spreadsheet and create columns for "Question," "Option A," "Option B," "Option C," and "Correct Answer."

• You can write 5-10 questions with multiple-choice answers and indicate the correct answer.

• You can use formatting to make it visually appealing (bold headings, different colours for options).

LAB ACTIVITY

COLOURFUL CHART CREATION

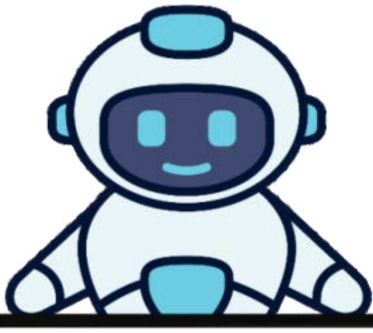


Create a colourful chart to represent data creatively.

- Provide students with a list of data (e.g., number of pets owned by classmates).
- Students enter the data into Excel and create a bar or pie chart.
- Encourage them to format the chart with colours and labels to enhance visual appeal.

LAB ACTIVITY

DAILY WEATHER LOG



Track daily weather conditions and create a visual representation.

- Create a spreadsheet with columns for "Date," "Temperature," "Weather Condition," and "Comments."
- Fill in this data for a week, describing the weather each day.
- You can then create a line chart to show temperature changes over the week.